Transfer learning and in-context learning for stage direction classification in French

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Introduction

Stage directions provide information complementing character speech, e.g. about performance or decoration, and can sometimes be independent from performance, targeting readers (Pfister 1988). They have been little studied using computational means (Dennerlein 2016, Maximova / Fischer 2019, Pagel et al. 2021). However, their automatic annotation on a large scale is useful, as they can give clues about genre stylistics and dramatic structure and technique.

This poster follows up on our previous work on stage direction classification in French via fine-tuning pre-trained language models and prompting large language models (LLM) (Schneider / Ruiz 2024). We expand the range of models, hyperparameters and prompts tested. Both paradigms provide useful results. We undertook a new qualitative analysis of LLM results, showing limits in our reference annotations, and how LLMs can help identify them.

Stage direction typology

The literature proposes several typologies, reviewed by Galleron (2021). Ours (see Schneider 2024 for details) includes 13 types (Table 1). We mapped 87 types appearing at least 50 times in FreDraCor's original annotation (Fischer et al. 2021, cf. Fièvre 2007) to our 13 types. Our types are intended to be useful for literary analysis and for testing classification models, as they represent different degrees of challenge for automatic classification; some categories are characterized by a clear vocabulary, some are more ambiguous.

Fine-tuning experiments (transfer learning)

Our earlier work fine-tuned several models, gradually reducing the number of examples to assess model efficiency. French monolingual models camembert-base (Martin et al. 2020) and distilcamembert-base (Delestre / Amar 2022) outperformed multilingual BERT (Devlin et al. 2019). We now test FlauBERT (French monolingual), whose base model outperforms CamemBERT's in certain domains (Le et al. 2020). We obtained better results than with CamemBERT, but only when fine-tuning on at least ca. 3,000 examples. We also tested xlm-roberta-base (Conneau et al. 2020), a more recent multilingual model, but could not outperform multilingual BERT. Decreasing learning rate or increasing batch size did not improve results. Thus, while it was relevant to fine-tune more recent models for comparison, CamemBERT models remain the best for the task according to our experiments, given their more stable performance when training data is reduced (Table 2 and Figure 1).

Prompting (in-context learning)

In (Ruiz / Schneider 2024), we tested gpt4o , gpt4o-mini and llama-3.1-8B with zero-shot and few-shot prompting; examples are always in French and we compared results with the rest of the prompt in English or in French. Here we additionally test mistral-large and mistral-small . Our original prompts included only one stage direction to classify; the rest of the prompt (category definitions, plus examples in few-shot mode) was repeated for each stage direction, which consumes many tokens. We now tested how many stage directions to classify we can include in a prompt without decreasing result quality. As Table 3 shows, results remained stable when classifying 75 stage directions with a single prompt. We should note however that with many stage directions in the same prompt, it is occasionally impossible to recover the category for a test-item from the model response, as the model may make formatting errors. We made our client and evaluation scripts more robust to handle this, and the huge savings (Table 5) in tokens (>90%) and response time (40-80%) outweigh this issue.

Qualitative analysis

The prompt asked the model to generate an explanation for each classification, and we used this to assist qualitative analysis, noticing several patterns:

- cases counted as LLM errors but that are due to inconsistencies in our reference annotations, e.g. cases of a character interacting with an object sometimes tagged as type *action* but sometimes as type *object* in the reference, whereas the model correctly assigns an *object* category. These cases suggest that LLM output can be used to refine our reference annotations.
- LLM errors that stem from incomplete treatment of some nuances by the model, e.g. stage directions related to non-verbal emotion expression like laughing or crying tagged by the model as a general *action* rather than a *delivery* stage direction. Such cases suggest the limits of prompting.

Our analyses suggest that fine-tuning excels at learning nuanced decision boundaries and will get a better F1 score (up to 0.09 higher than in-context learning, see Tables 2-4) even with a conceptually flawed typology. Conversely, prompting, with the more limited knowledge about the typology derived from the context given to it, may draw our attention to such flaws and may be used to help improve; bad performance for a category may indicate reference annotation errors and explanations generated by the LLM can help detect error patterns.

Conclusion

The results have practical value regarding stage direction classification via fine-tuning or prompting, and inform next steps regarding an automatic annotation task useful for stylistic and genre analysis.

Once we refine the methods whereby we produce our reference categories, it should be possible to improve classification results and go beyond a classification task. For instance, it could be tested whether LLMs identify characters participating in the event represented by the stage direction (cf. Galleron 2021). Applying a solid annotation schema would be required before producing reference data to evaluate stage direction classification and related tasks multilingually.

Code and data availability

Publicly available at https://github.com/pruizf/d25-llm-stdir

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
gpt-4o	1	3,259,042		6,053.86		0	6,985,527		2,413.29		0
gpt-4o	75	217,120	93.34	1,552.17	74.36	0	147,808	97.88	436.84	81.90	0
mistral-small	1	4,278,112		4,502.77		0	8,687,014		1,828.52		0
mistral-small	75	282,212	93.40	2,460.57	45.35	6	190,071	97.81	373.54	79.57	10
mistral-large	1	4,258,542		8,783.03		0	8,667,889		2,078.10		0
mistral-large	75	292,144	93.14	5,203.56	40.75	0	185,284	97.86	736.64	64.55	0
testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Table 1. Stage direction typology. Details in Schneider (2024), Schneider / Ruiz (2024).

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
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testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Table 2. Fine-tuning pre-trained language models. Macro-F1 score (average over 5 runs) when fine-tuning with a gradually decreasing number of examples, testing always on 2,923 examples. Classical machine learning methods are shown as baseline. Best model score at each proportion of examples is bolded. Models marked with (+) were added in this study.

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
gpt-4o	1	3,259,042		6,053.86		0	6,985,527		2,413.29		0
gpt-4o	75	217,120	93.34	1,552.17	74.36	0	147,808	97.88	436.84	81.90	0
mistral-small	1	4,278,112		4,502.77		0	8,687,014		1,828.52		0
mistral-small	75	282,212	93.40	2,460.57	45.35	6	190,071	97.81	373.54	79.57	10
mistral-large	1	4,258,542		8,783.03		0	8,667,889		2,078.10		0
mistral-large	75	292,144	93.14	5,203.56	40.75	0	185,284	97.86	736.64	64.55	0
testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Figure 1. Fine-tuning results as training set is reduced. Models with (+) in the legend were added in this study.

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
gpt-4o	1	3,259,042		6,053.86		0	6,985,527		2,413.29		0
gpt-4o	75	217,120	93.34	1,552.17	74.36	0	147,808	97.88	436.84	81.90	0
mistral-small	1	4,278,112		4,502.77		0	8,687,014		1,828.52		0
mistral-small	75	282,212	93.40	2,460.57	45.35	6	190,071	97.81	373.54	79.57	10
mistral-large	1	4,258,542		8,783.03		0	8,667,889		2,078.10		0
mistral-large	75	292,144	93.14	5,203.56	40.75	0	185,284	97.86	736.64	64.55	0
testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Table 3. Prompting results with a single stage direction to classify in the prompt vs. 75 stage directions, in zero-shot or few-shot mode (20 examples per class). Best results per mode bolded. Examples are always in French; fr and en are the language in the rest of the prompt.

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
gpt-4o	1	3,259,042		6,053.86		0	6,985,527		2,413.29		0
gpt-4o	75	217,120	93.34	1,552.17	74.36	0	147,808	97.88	436.84	81.90	0
mistral-small	1	4,278,112		4,502.77		0	8,687,014		1,828.52		0
mistral-small	75	282,212	93.40	2,460.57	45.35	6	190,071	97.81	373.54	79.57	10
mistral-large	1	4,258,542		8,783.03		0	8,667,889		2,078.10		0
mistral-large	75	292,144	93.14	5,203.56	40.75	0	185,284	97.86	736.64	64.55	0
testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Table 4. Per category results for the best performing models under fine-tuning (transfer learning) and prompting (in-context-learning). Fine-tuning was done with 100% of the available examples (9,352 training + 2,338 validation). N: testset size for each category.

model	stg dir to classify per prompt	zero-shot					few-shot				
		total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir	total tokens	% tokens saved	response time (sec)	% sec saved	lost stg dir
gpt-4o-mini	1	3,262,985		4,511.95		0	6,983,531		1,340.31		0
gpt-4o-mini	75	217,384	93.34	1,285.97	71.50	5	146,167	97.91	311.51	76.76	0
gpt-4o	1	3,259,042		6,053.86		0	6,985,527		2,413.29		0
gpt-4o	75	217,120	93.34	1,552.17	74.36	0	147,808	97.88	436.84	81.90	0
mistral-small	1	4,278,112		4,502.77		0	8,687,014		1,828.52		0
mistral-small	75	282,212	93.40	2,460.57	45.35	6	190,071	97.81	373.54	79.57	10
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mistral-large	75	292,144	93.14	5,203.56	40.75	0	185,284	97.86	736.64	64.55	0
testset size		2,923					877				
examples per prompt (as context)		0					260 (20 per class)				

Table 5. Token count and response times with French prompts, classifying a single stage direction per prompt vs. 75 stage directions per prompt. Token and response time savings in the latter setup provided. Column lost stg dir is the number of cases where, given formatting errors, the category for a stage direction cannot be recovered from the model response.

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